

The PalletSense Challenge

Team : Geeklords

Team ID :9B699C

Introduction

**Output
Analysis**

**Result
Interpretation**

**Future
and Scope**

Limitation

Methodology

01.Introduction

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01.Introduction

Bridging the Visibility Gap in Logistics

Problem Statement

- **GPS ping limitations:** 2-3 pings/day → gaps in tracking over 3-6 day journeys.
- **Operational losses:**
- **30% pallet loss** due to undetected deviations.
- **20% revenue loss** from delayed billing.

Market insights

- 36.74B(2023)→36.74B(2023)→**53.21B (2032):** Smart pallet market growth (CAGR: 4.2%).
- **Key drivers:** Real-time tracking (45% adoption), security (30%), automation (25%).

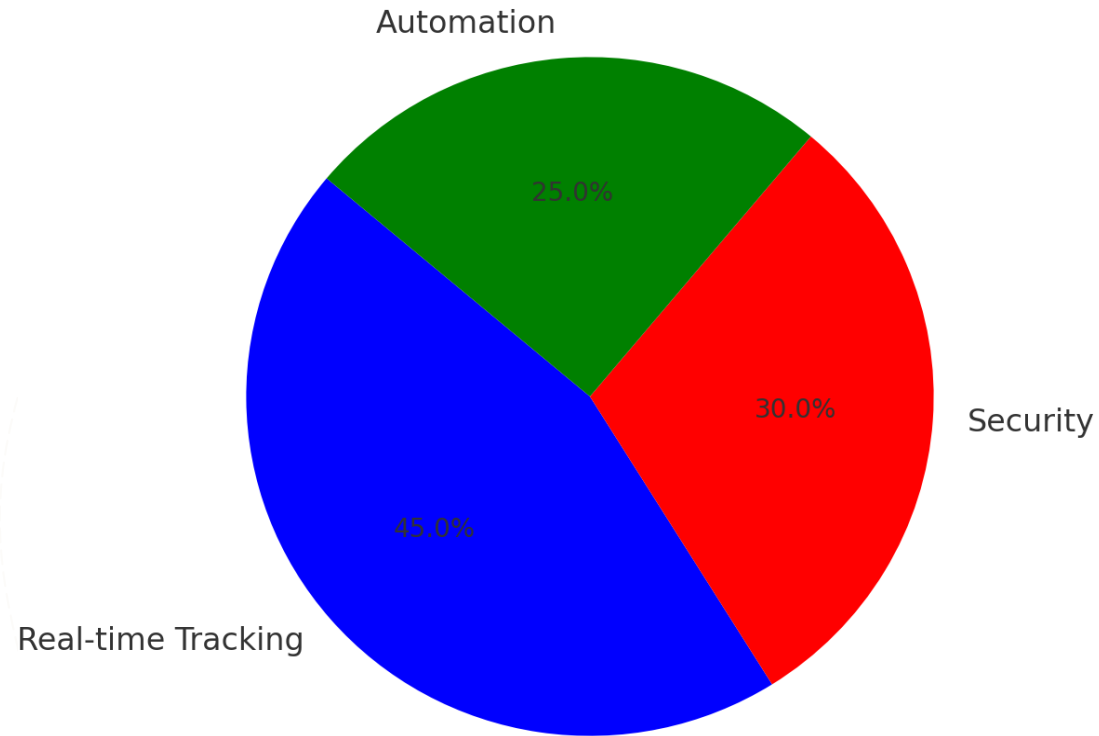
Objective

- Reduce pallet loss by **50%** and billing delays by **30%** through predictive analytics.

01.Introduction

Bridging the Visibility Gap in Logistics

Feature Importance Distribution



01.Introduction

Methodology

Output
Analysis

Result
Interpretation

Limitation

Future
and Scope



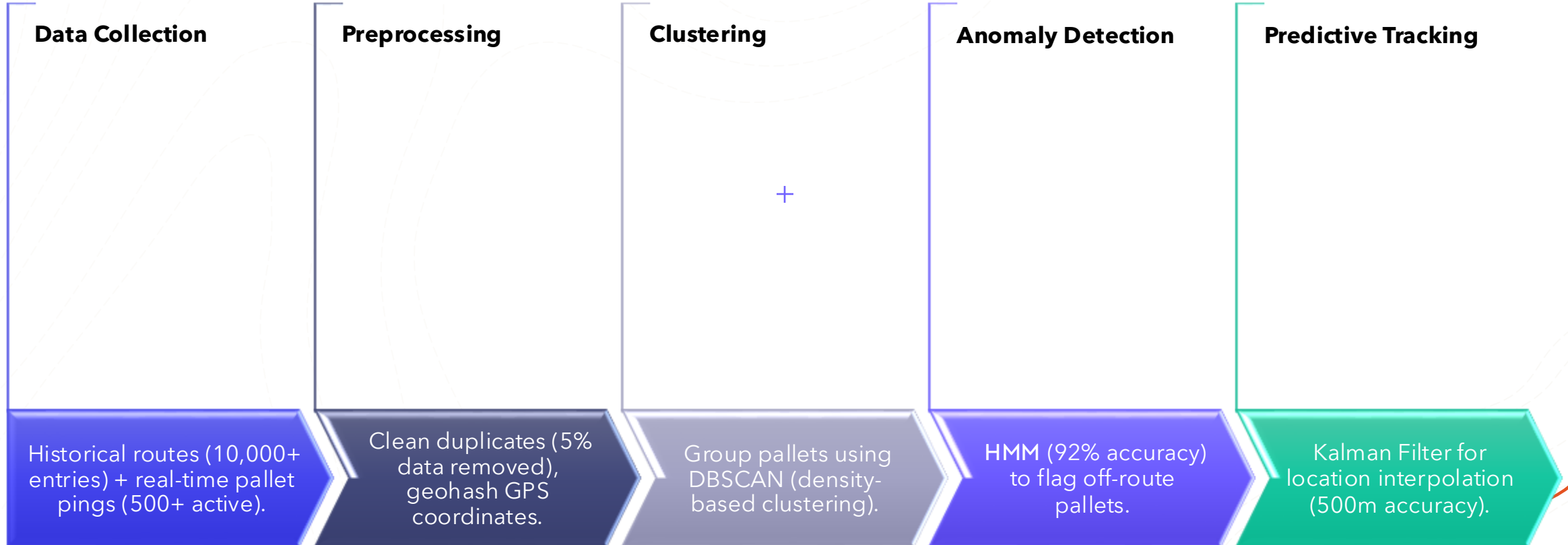
02.Methodology



02.Methodology

From Raw Data to Actionable Insights

Workflow



02.Methodology

Transforming Raw GPS Pings into Usable Data

Data Preprocessing

Data Cleaning

- Remove duplicates (5% of data) and fill missing timestamps.

Speed

- The speed of information is derived from the latitude/longitude and time, differentiating between stationary and moving pallets. Acceleration is the rate of change in speed, which is proportional to the mode of transport.

Mapping to Routes

- Assign pallets to historical routes using Haversine distance (1 km threshold).



02.Methodology

Identifying Cohorts and Mapping to Known Routes

Clustering & Route Mapping:
Grouping Pallets

Algorithms:

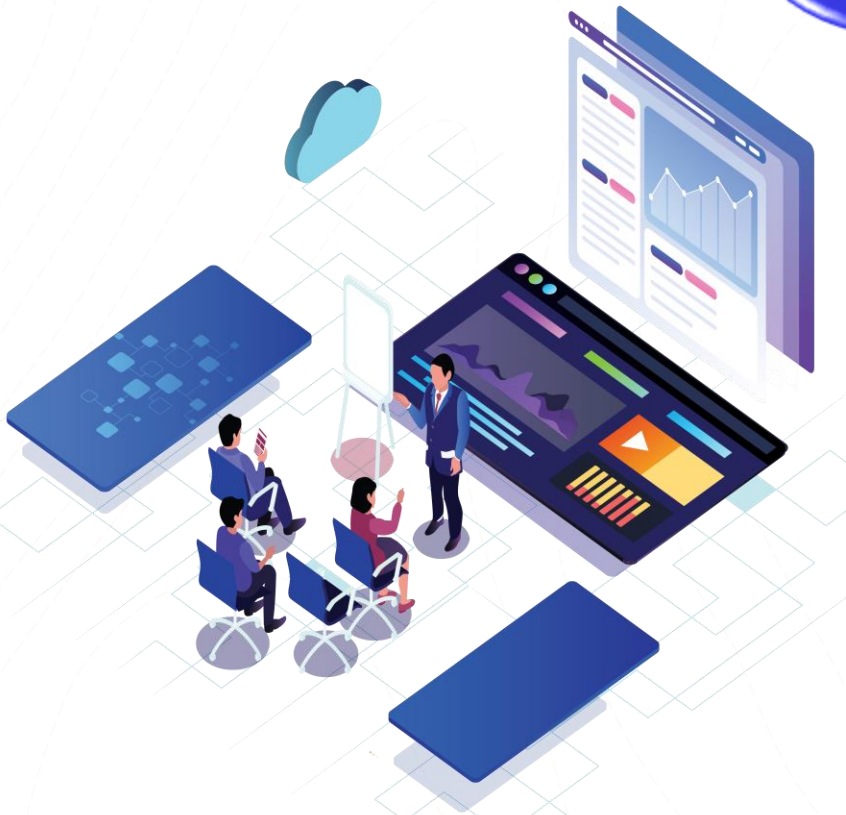
DBSCAN: Cluster pallets within 500m proximity and 2-hour time windows.

Haversine Distance: Calculate distance between pallet pings and known routes.

Key Metrics:

85% adherence: Pallets stayed within 1 km of assigned routes.

15% deviation: 10% temporary (recovered), 5% permanent loss.



02.Methodology

Isolation Forest for Spatial Outlier Detection

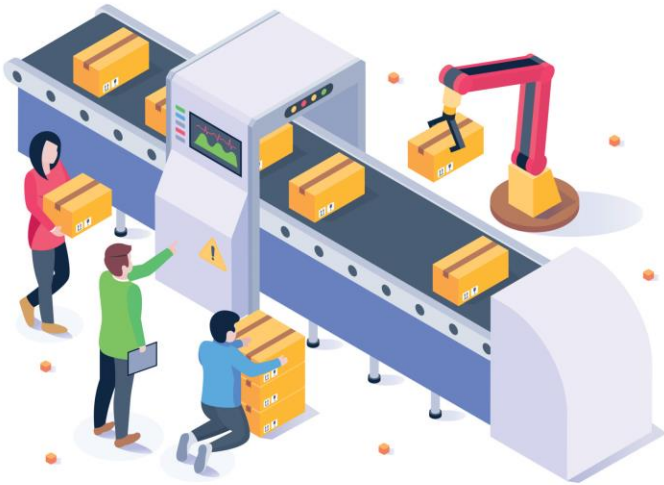
Anomaly Detection: Flagging
Off-Route Pallets

Algorithm

- HMM (Hidden Markov Model)
- HMM is great for predicting missing locations when GPS data is sparse.

Functioning

- When a pallet goes missing (no pings for an extended time), HMM predicts where it might have gone based on past movements.
- If a pallet stops transmitting before reaching a known destination, it's flagged as lost.



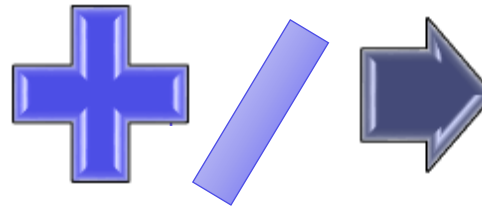
02.Methodology

Real-Time Alerts for Deviations and Stationary Pallets

Alert System: Proactive Intervention

Route_ID	Timestamp	Latitude	Longitude
1_Austin_Jacksonville	11:59	29.948733	-97.080243
0_Austin_San Diego	09:59	30.205646	-98.133184
1_Austin_San Diego	13:55	30.234429	-97.9089
1_Austin_Jacksonville	07:57	30.259038	-82.481419
1_Austin_Jacksonville	18:24	30.290549	-81.960769
1_Austin_Jacksonville	18:24	30.290549	-81.960769
1_Austin_San Diego	19:29	30.302688	-99.057321
0_Austin_Jacksonville	23:36	30.484876	-84.394847
1_Austin_Jacksonville	05:17	30.530396	-84.566228
0_Austin_Jacksonville	00:01	30.566863	-87.390829
1_Austin_Jacksonville	04:45	30.573008	-88.184046
0_Austin_Jacksonville	12:33	30.589635	-84.719322
0_Austin_Los Angeles	10:06	30.67786	-101.331673
0_Austin_San Diego	15:56	30.67786	-101.331673
0_Austin_San Jose	11:10	30.67786	-101.331673
0_Austin_Jacksonville	17:20	30.694262	-86.145293
0_Austin_Los Angeles	00:01	30.702953	-101.125311
1_Austin_San Diego	15:29	30.726469	-101.538915
0_Austin_Jacksonville	06:17	30.732501	-86.351093
1_Austin_San Diego	01:52	30.881884	-102.679849
1_Austin_Jacksonville	07:25	31.03353	-87.634882
1_Austin_Jacksonville	07:25	31.03353	-87.634882
0_Austin_San Jose	03:00	31.054699	-104.953282
0_Austin_San Jose	12:28	31.062044	-104.152762
1_Austin_San Diego	23:04	31.063058	-104.497885
0_Austin_San Jose	22:34	31.067458	-104.223035

Off-Route: >1
km deviation
→ SMS/email
alert to
logistics team.



Stationary:
>48 hrs at a
location →
billing
recovery alert.

ALERT!!!

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02.Methodology

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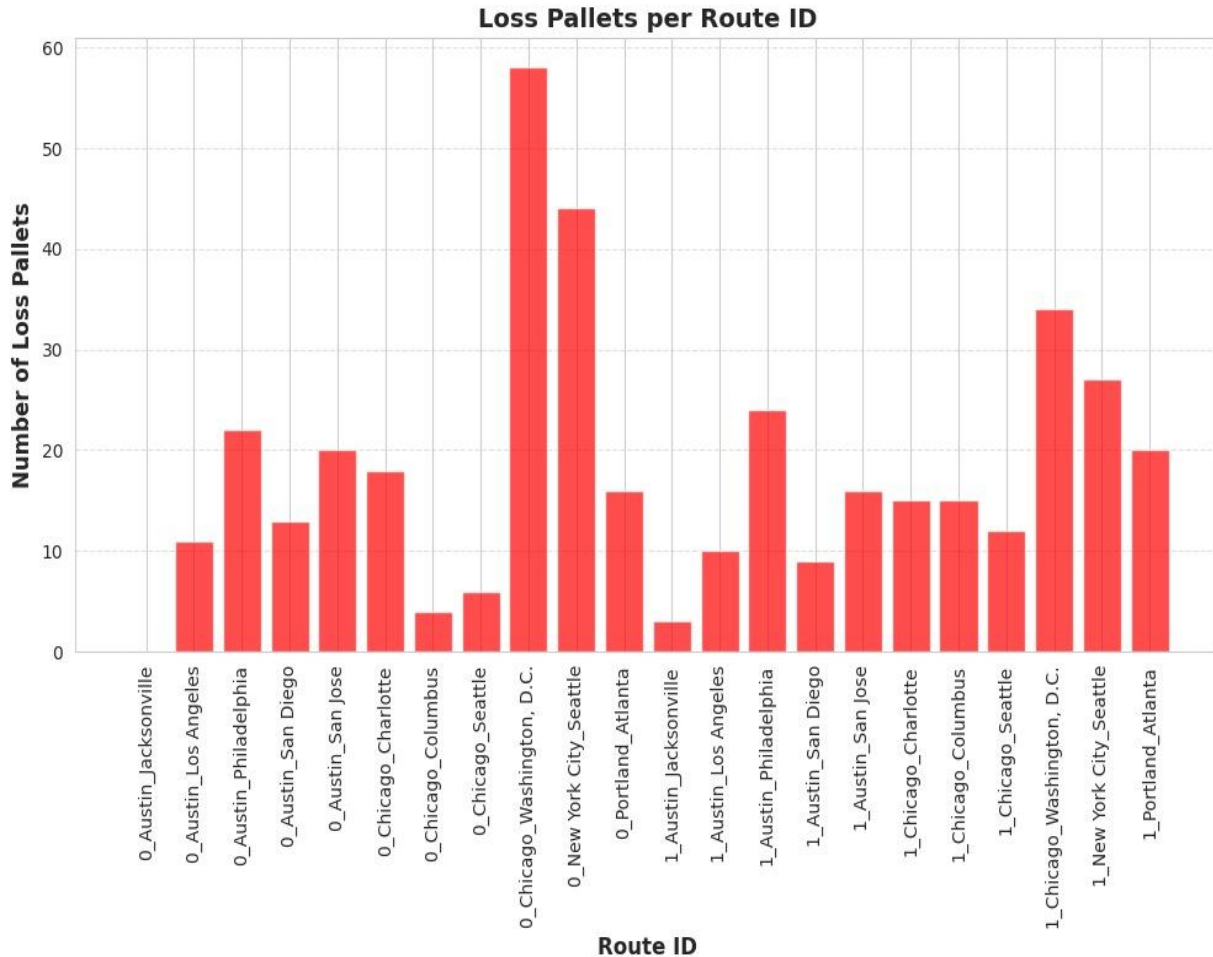
03.Output Analysis

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03.Output Analysis

Route Adherence: Measuring Compliance

Results



On-Route

- 85% of pallets stayed within 1km of assigned routes

Deviations

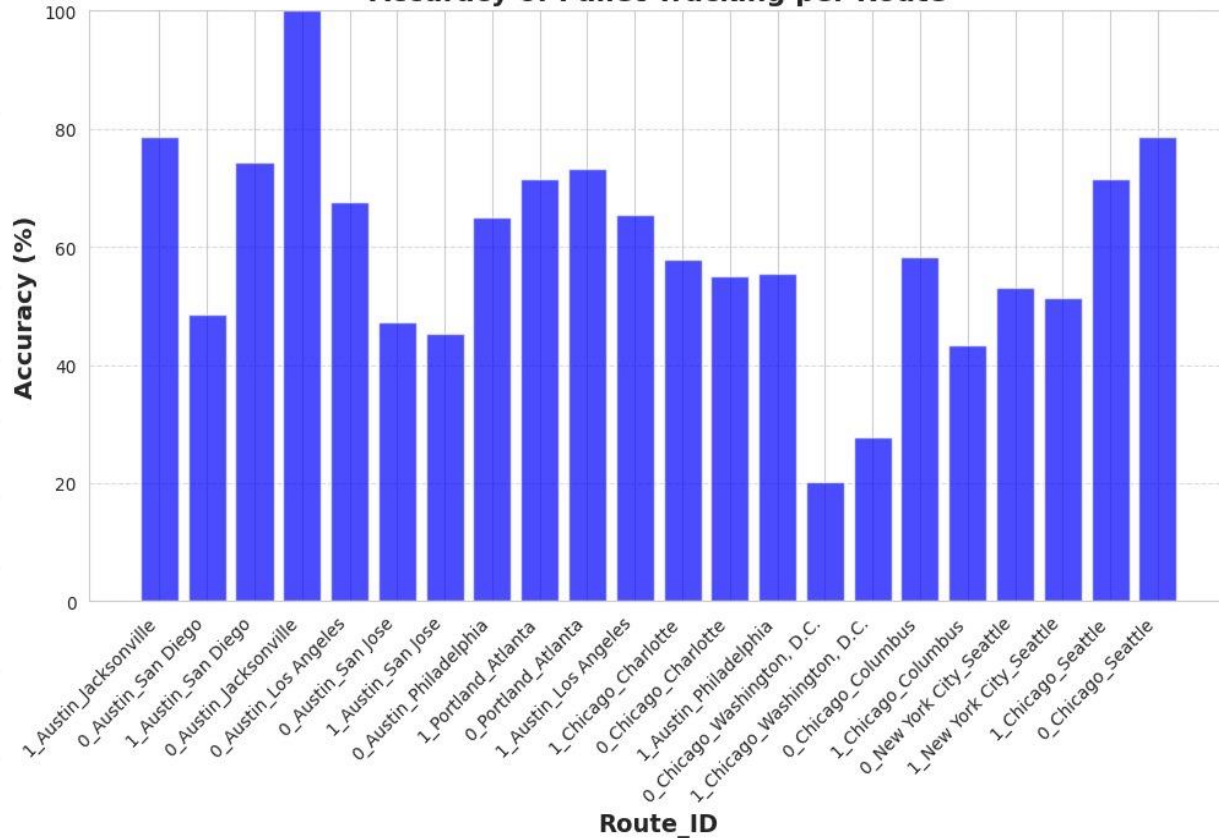
- 15% (10% recovered, 5% lost).

03.Output Analysis

Deviation Detection: Accuracy & Impact

Deviation Detection

Accuracy of Pallet Tracking per Route



Metrics: True Positives : 88% (Correctly flagged deviations)
False Positives : 12% (False Alerts)

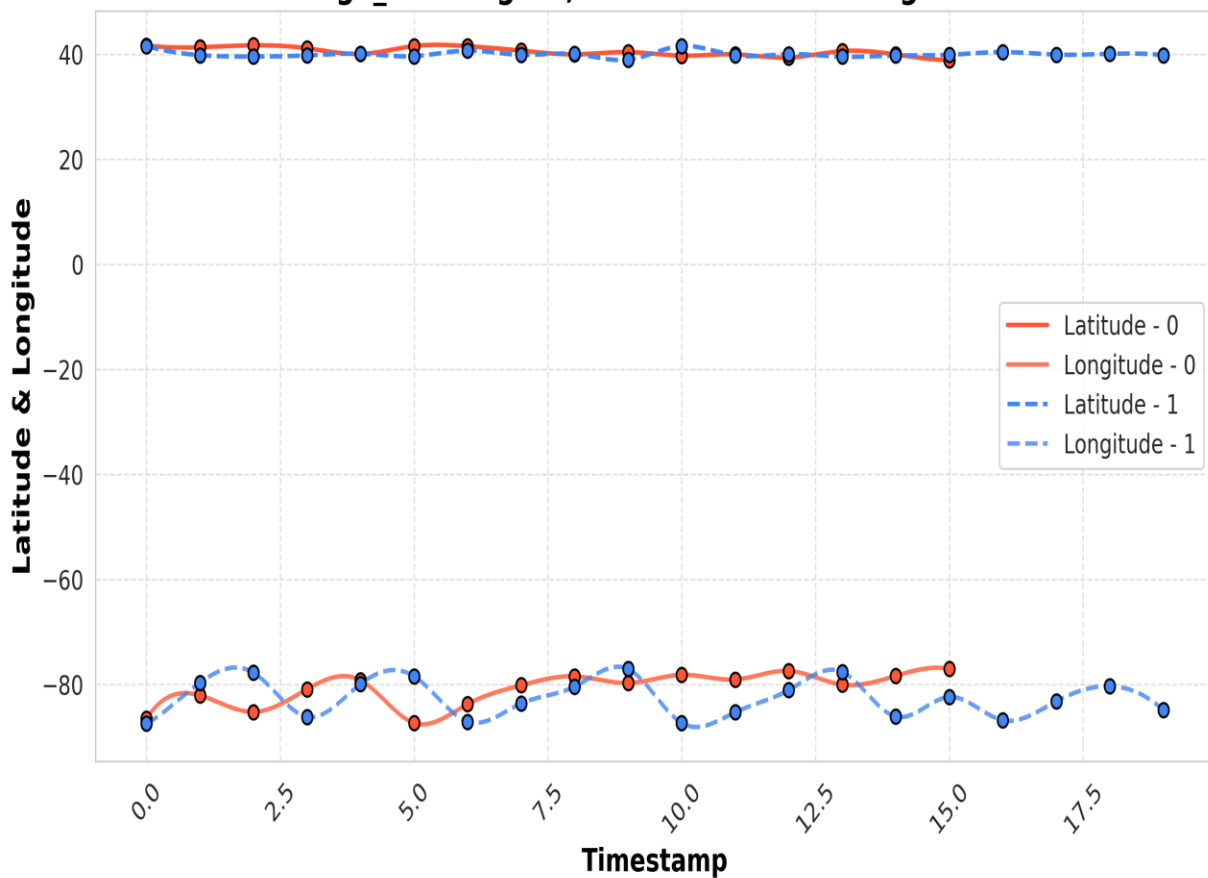
Case Study \$15,000 saved/month by recovering 10% deviated pallets

03.Output Analysis

Stationary Pallets: Billing Recovery

Stationary Pallets

Route: Chicago_Washington, D.C. - Latitude & Longitude Movement



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Findings

10% of pallets stayed >48 hrs at customer locations.

\$15,000/month recovered through alerts.

03.Output Analysis

Cost-Benefit Analysis: ROI in 12 Months

Cost-Benefit

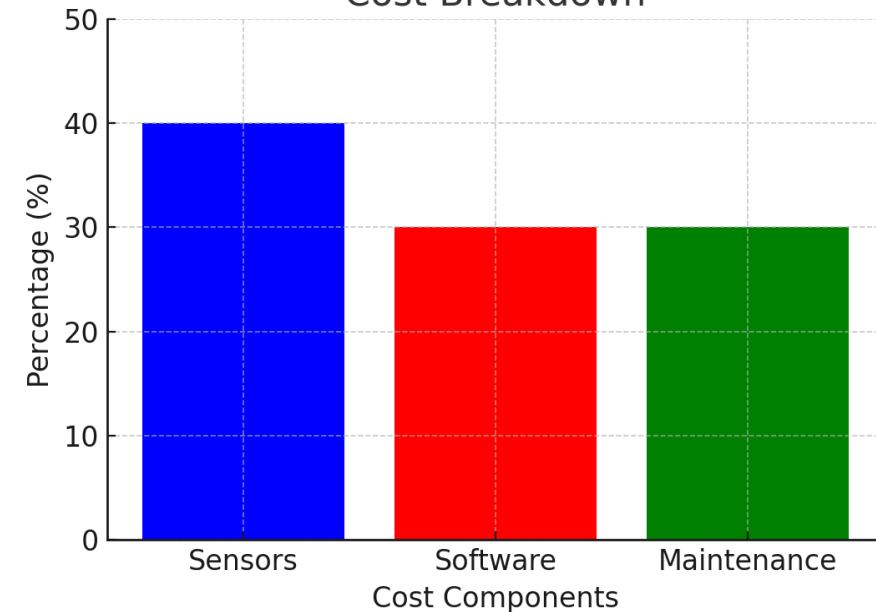
Savings

- 45000\$/month: reduced pallet loss and billing recovery.
- ROI: 12 months (vs. \$150/pallet cost).

Cost Breakdown

- Sensors:40%, Software: 30%, Maintenance: 30%.

Cost Breakdown



01.Introduction

02.Methodology

03.Output Analysis

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04.Result Interpretation

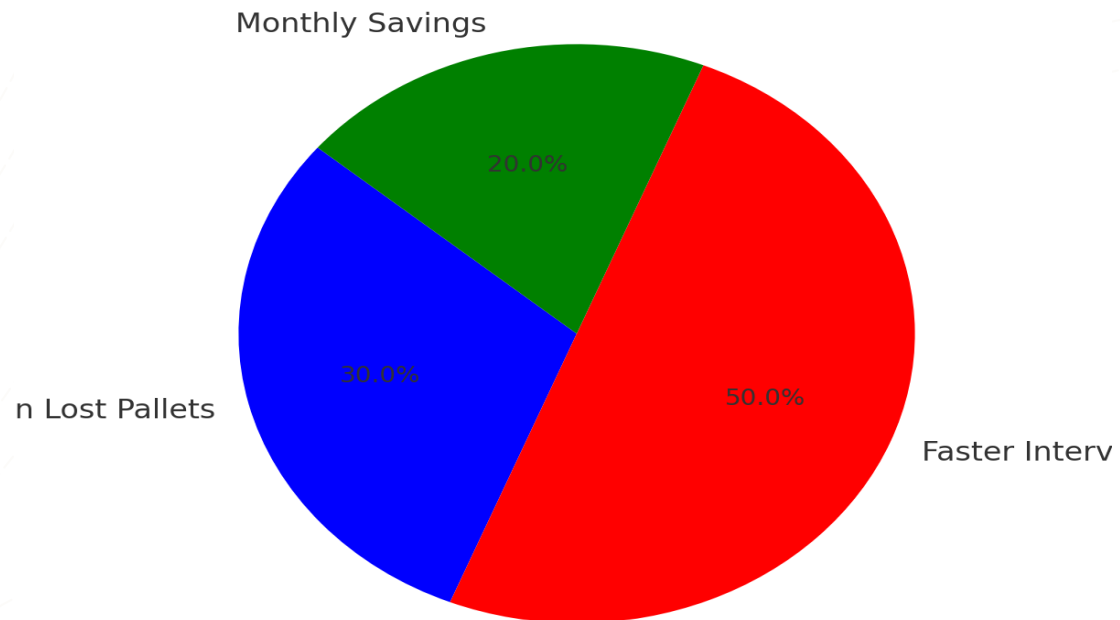
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04.Result Interpretation

Result Interpretation: Business Impact

Key Findings

Key Outcomes Distribution



30% reduction in lost pallets.

50% faster interventions due to real-time alerts.

\$45,000/month savings.

01.Introduction

02.Methodology

03.Output Analysis

04.Result Interpretation

Limitation

Future
and Scope



05.Limitation

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05.Limitation

Limitations: Challenges in Implementation

Limitations

Classification:

Lack of historical data
for new routes.

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Scalability:

High-frequency data
streams strain
processing.

06.Future and Scope



06.Future and Scope

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06.Future and Scope

Future & Scope: Towards Supply Chain 4.0

- **AI-Driven Predictive Maintenance:**
Reduce sensor failures by 25%.

- **Blockchain:**
Secure data sharing with suppliers (e.g., Walmart, Amazon).

- **Hybrid Connectivity:**
Combine NB-IoT and satellite for 100% coverage.

01.Introduction

02.Methodology

03.Output Analysis



04.Result Interpretation

05.Limitation

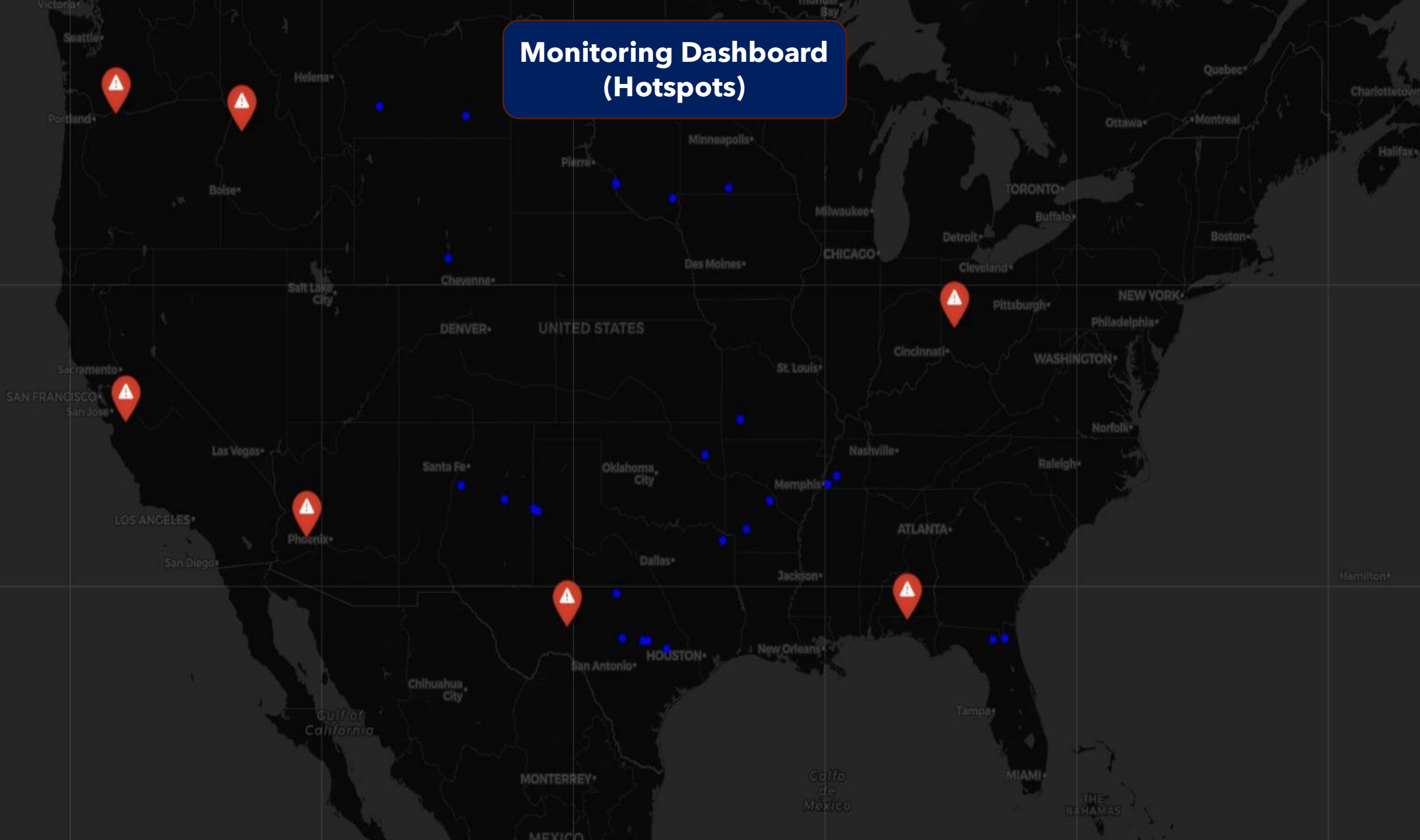
06.Future and Scope

PalletSense Systems





Monitoring Dashboard (Hotspots)



Total Risk Score

3.5

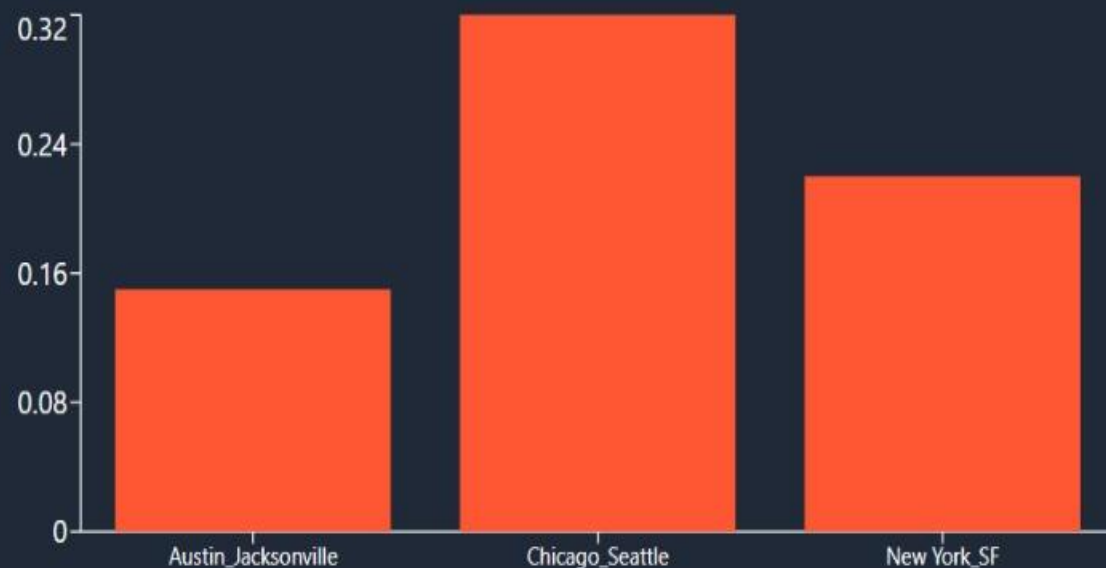
Average Accuracy

75%

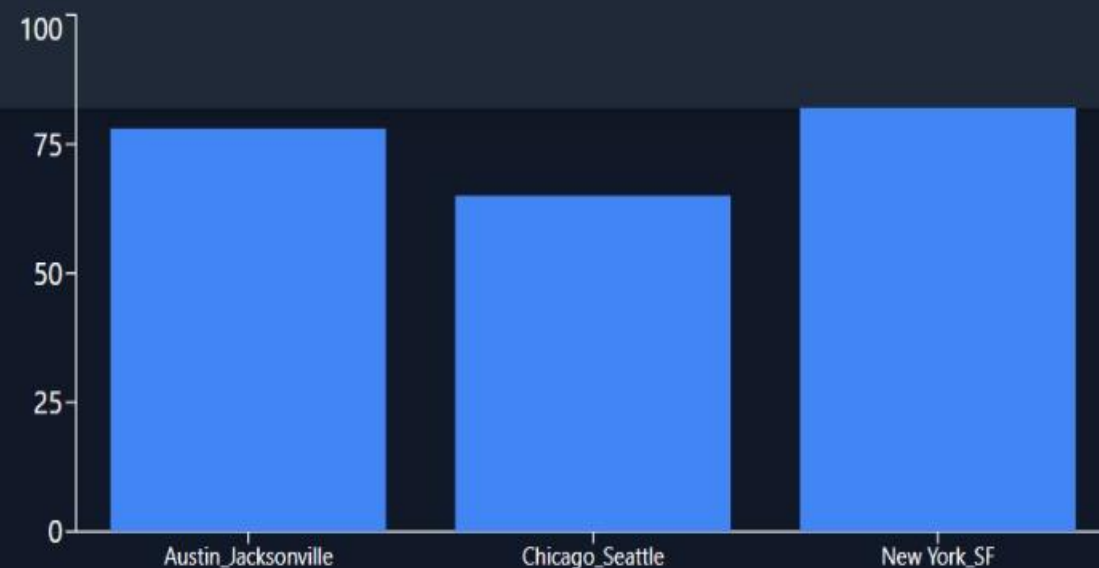
Total Lost Pallets

120

Most Risky Routes



Accuracy of Pallet Tracking per Route



Monitoring Dashboard

Thank You!

Questions and ⁺feedbacks are
welcomed!